

ASHISH VERMA

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EDUCATION

Indian Institute of Technology Guwahati (IITG)
Master of Technology in Computer Science and Engineering

2024 - 2026

Kamla Nehru Institute of Technology Sultanpur (KNIT)
Bachelor of Engineering in Computer Engineering

2018 - 2022

PROJECTS

Design a PEM Algorithm for the LCS Problem | MTech Thesis Project July 2025 – Present

- **Designing** Parallel Algorithms for the **Longest Common Subsequence (LCS)** problem using **Parallel External Memory (PEM)**, aiming to speed up processing and reduce I/O overhead for very large data.

Logic Synthesis Using Heuristics |  Feb. 2025 – Apr. 2025

- **Designed** a neural network combining **GCN and embedders** to predict and optimize circuit delay in different scaled AIGs, cutting logic synthesis **time from 3 hours (manual) to 20 minutes** with **$\pm 4.02\%$ error**.
- **Tech stack:** Python, PyTorch, Shell Scripts, ABC tool, Few-Shot Learning, Simulated Annealing, Monte Carlo Optimization.

Voice Based Dictionary Search |  Oct. 2024 – Nov. 2024

- **Developed a speech recognition system** that recognizes spoken words in real time and displays their meanings as part of a dictionary search tool, achieving an **88% accuracy** using **Hidden Markov Models**.
- **Trained** the system using pre-recorded audio datasets to handle various accents and pronunciations effectively.
- **Tech stack:** C++, Hidden Markov Models, Levinson-Durbin algorithm, LBG clustering.

Blog-Bar - Full-Stack Blog Platform | *Sept. 2025*

- A **Full-stack** blogging platform designed for creating, sharing, and managing articles. It provides a secure environment for authors to publish content and a **dynamic, searchable** experience for readers.
- Engineered a secure **Node.js** API for content management and designed a responsive **React** front-end with features like a rich text editor and an admin dashboard.
- **Tech Stack:** React, Redux, Node.js, Express.js, MongoDB, JWT, and Tailwind CSS.

Secure Hybrid TCP/UDP Client-Server Framework | Apr. 2025 - May 2025

- Designed and implemented a **secure client-server communication system** in C++ to demonstrate and compare **FCFS** and single-exchange **Round Robin (RR)** server-side scheduling policies.
- Engineered a **hybrid TCP/UDP architecture**, using TCP for reliable session initialization and dynamic UDP port allocation, followed by high-speed UDP-based encrypted data transfer.
- Developed a custom **16-byte symmetric XOR encryption scheme** combined with **Base64 encoding** to ensure secure and transmission-safe communication.
- Implemented configurable **server scheduling strategies**, throughput measurement client, and a **debug mode** for inspecting encrypted payloads.
- **Tech Stack:** C++, Socket Programming, TCP/UDP Protocols, Symmetric Encryption, Base64 Encoding.

ACHIEVEMENTS

- **AIR 733**, GATE CSE (**99.40 %ile**) among **123,967** students
- **Competitive Programming**, Codeforces **Specialist** with a contest rating of **1406**
- **Coding**, Solved **900+** **Leetcode** Problems honing Data Structures and Algorithms(DSA)

SKILLS

- **Languages:** C/C++, Python
- **Web Technologies:** HTML, CSS, JavaScript, React, Node.js, Express.js
- **Databases:** MongoDB, MySQL
- **Tools:** Git, GitHub, Linux (Elementary), Google Colab, Visual Studio, Windows

RELEVANT COURSEWORK

- Design & Analysis of Algorithm
- FM for System Verification
- Speech Processing
- Introduction to ML (NPTEL)
- ML for EDA
- Data Structures Lab
- High Performance Computing
- Cloud Computing (NPTEL)
- Systems Lab